PBEBench: A Multi-Step Programming by Examples Reasoning Benchmark inspired by Historical Linguistics

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Abstract

Recently long chain of thought (LCoT) Large Language Models (LLMs) took the machine learning world by storm with their reasoning capabilities. However are the abstract reasoning abilities of these models good enough for problems of practical importance? Unlike past work which has mainly focused on math, coding, and data wrangling, we focus on a historical linguistics-inspired reasoning problem, formulated as Programming by Examples.

First-year students of historical linguistics are often asked to solve problems in which they are given a list of sets of etymologically related words from a few related modern languages (cognate sets) and asked to infer (1) a set of reconstructed words from the hypothetical most recent common ancestor of the languages and (2) chronologically ordered lists of string-rewrite rules, one for each language, by which each of the modern words can be derived from the reconstructed words. A good deal of progress has been made towards implementing (1), which can be seen as a sequence transduction problem (XYZ). The logic behind (2) is actually quite straightforward, but it proves challenging for contemporary ML models.

Consider the case of Huishu, a Tangkhulic language of Northeastern India. In this language, Proto-Tangkhulic * became u everywhere (Rule A: > u) and Proto-Tangkhulic *u become uk at the end of words (Rule B: u# > uk#). Proto-Tankhulic *uk became u everywhere (Rule C: uk > u). Rule A must have applied before Rule B. Otherwise words like *n ‘laugh’ would have become nu rather than the attestwed nuk in Huishu. Linguists say that Rule A FEEDS Rule B (creates a context where it can apply). Likewise, Rule C must apply before Rule B. Otherwise, words like *ru would become ru in Huishu, rater than the attested ruk. Linguists could say that Rule B COUNTER-FEEDS Rule C (if the order of the two processes were reversed, Rule B would create contexts where Rule C could apply, that is, B would feed C). The reverse of feeding is BLEEDING, in which a rule D destroys a context win which a rule E would apply. It has a counterfactual version, COUNTER-BLEEDING (in which Rule E would bleed Rule D if it applied first). In the history of a language, if the order of two sound changes matters, one of these relations holds between them.

More generally, since counter-feeding and counter-bleeding are simply counterfactual versions of feeding and bleeding, the chronologies of sound changes can be reasoned about by students of historical linguistics simply in terms of the potential feeding and bleeding relations between them. Since these relations can be defined algorithmically (see Section XYZ), the underlying logic is simple. The forward reconstruction problem simply involves defining the rules jointly with a multi-step plan consistent with that logic.

This “forward reconstruction” task, which combines inductive reasoning and planning, is simply a specific instance of a more general kind of reasoning problem, which has been a topic of discussion since the early years of artificial intelligence. It is related to all problems where it is necessary to

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plan a discrete sequence of processes P that will transform the elements of a source set S to the corresponding elements of the target set T. Problems of this type appear in domains as varied as computer science and materials science.

The “forward reconstruction” formulation of the problem, though, has a number of properties that make it particularly attractive for benchmarking language models: (1) It can be structured to largely domain-specific biases—so that it is strictly a reasoning task. (2) The reasoning necessary to complete the task can be expressed via a very simple pair of logical relations between processes (feeding and bleeding). (3) The processes themselves can be expressed very simply since they are, logically, just pairs for the form $\langle a, a' \rangle$ (or, alternatively, rewrite rules of the form $a \rightarrow a'$. These can be represented trivially in Python code.

1 Related Work

Program synthesis and Programming By Example Early symbolic techniques searched within a hand-crafted DSL using enumeration or constraint solving Summers [1977], while domain-tailored systems such as FlashFill Gulwani [2011] delivered practical scale through clever ranking over string-manipulation programs. More recently, neural models cast synthesis as sequence translation from a specificationnatural language or I/O pairsto code Maddison and Tarlow [2014], Balog et al. [2017], Rabinovich et al. [2017]. Symbolic approaches remain brittle outside their DSL, and neural ones falter with scarce data or intricate control flow. We target PBE tasks whose rules mirror rich linguistic structure, a regime that stresses both families.

LLMs for Program Synthesis Chen et al. [2021], Li et al. [2022], and successors achieve near-human performance on competitive programming by mapping prose to code. Yet when outputs must strictly obey a grammar, LLMs frequently err; grammar-prompting mitigates this by injecting BNF constraints Wang et al. [2023]. In PBE settings, LLMs exploit in-context examples Brown et al. [2020] but remain sensitive to prompt phrasing and rarely generalize systematically. Our experiments show that linguistically inspired PBE amplifies these weaknesses and motivate structural scaffolds.

Probing LLM understanding of linguistic structure. Probes test whether LLMs internalize syntax, morphology, and phonology Linzen et al. [2016], Gulordava et al. [2018], Warstadt et al. [2019]. Formal-language studies link learnability to language rank and string length Borenstein and et al. [2024], while compositional splits such as SCAN reveal pattern-matching rather than rule abstraction McCoy et al. [2019]. These results foreshadow the difficulty of inferring phonological or hierarchical rewrite programs from scant examplesthe core demand of our benchmark.

LLM benchmarks SyGuS Alur et al. [2013] and FlashFill datasets measure DSL-bounded synthesis; HumanEval Chen et al. [2021] and MBPP Austin et al. [2021] test NL-to-code generation; GSM8K Cobbe et al. [2021] and ARC Clark et al. [2018] probe reasoning; SCAN and its variants target systematic generalization Lake and Baroni [2018], Keysers et al. [2020]. None, however, evaluate LLMs on *linguistically driven* PBE where models must induce executable rules that manipulate hierarchical structure from sparse examples. We fill this gap with a new dataset and protocol.

2 Theoretical Framework

2.1 Feeding

Here is a proposed algorithm for feeding:

$$\text{feeds}(a \to a', b \to b') = \begin{cases} \bot & a = a' \\ \top & a' = \epsilon \wedge |b| > 1 \\ \top & a' \in \mathsf{Substr}(b) \wedge \mathsf{Count}(a', a) > 1 \\ \top & b \in \mathsf{Substr}(a') \wedge \mathsf{Count}(b, a') > \mathsf{Count}(b, a) \\ \top & |\mathsf{OverlapDiff}(a, a', b)| > 0 \\ \bot & \text{otherwise} \end{cases} \tag{1}$$

where $\mathsf{Pref}(s)$, $\mathsf{Suff}(s)$, and $\mathsf{Substr}(s)$ are the sets of prefixes, suffixes, and substrings of s, respectively, $\mathsf{Count}(s, t)$ is the count of unique occurrences of s in t, $\mathsf{Overlap}(s, t)$ is $(\mathsf{Pref}(s) \cap \mathsf{Suff}(t)) \cup (\mathsf{Suff}(s) \cap \mathsf{Pref}(t))$, and $\mathsf{OverlapDiff}(s, t, u)$

$$\text{bleeds}(a \to a', b \to b') = \text{feeds}(a' \to a, b \to b') \tag{2}$$

Where $B(\cdot)$ is a function that returns all substrings, $S(\cdot)$ is a function that returns all suffixes and $P(\cdot)$ is a function that returns all prefixes.

Definition 2.1 (structural description)**.** The *structural description* of a rule $a \to a'$ is met by each case in which a occurs as a substring of Σ.

Definition 2.2 (feed)**.** A rule $A = a \to a'$ *feeds* a rule $B = b \to b'$ if there exists some string $\sigma \in \Sigma^*$ such at applying A to σ yields a string τ such that the structural description of B is satisfied by τ more times than it was satisfied by σ.

Definition 2.3 (overlap)**.** Two strings a and b will be said to *overlap* if there is a non-empty prefix of a that is a suffix of b or a non-empty prefix of b than is a suffix of a.

Lemma 1. *Any rule of the form $a \to \epsilon$ feeds any rule of the form $b_{1:m}b_{m+1:n}$*

Proof. Let rules A and B be string substitution rules of the form $a \to \epsilon$ and $b_{1:m}b_{m:n} \to b'$. In Σ^* there exist, by definition of Kleene closure, an infinite number of strings of the form $b_{1:m}a_{:}b_{m:n}$. By definition of string substitution, there exists some $\sigma \in \Sigma^*$ such that the application of A to σ yields subtrings $b_{1:m}b_{m:n}$ that did not exist in σ □

Lemma 2. *Any rule of the form $a \to a'$ feeds a rule of the form $b \to b'$ if b is a substring of a' and not of a.*

Proof. Let rules A and B be string substitutions rules of the form $a \to a'_{:n}ba'_{n:}$ and $b \to b'$ and and that b is not a substring of a and that A does not feed B. By the definition of *structural description*, this would entail that there strings of the form $a'_{:n}ba'_{n:}$ which do not include b, which would be a contradiction □

AN: And we also discussed a way to potentially combine 2,3 and 4: Lemma 1: "If a' is a subsequence of a and substring of b but a is not a substring of b"

DRM: Can you provide examples of how this works in each case?

Lemma 3. *Any rule A of the form $a \to xy$ feeds any rule B of the form $zx \to b'$ if x is not a substring of a.*

Proof. Suppose that there are two rules $A = a \to xy$ and $B = zx \to b'$. If we assume that A does not feed B when x is not a substring of a, it follows that there are no strings in Σ^* of the shape zxy for which x is not a substring of a. Since, by definition, Σ^* contains all strings over the alphabet Σ, by the definition of the Kleene closure, this would be a contradiction □

Lemma 4. *Any rule A of the form $a \to yx$ feeds any rule B of the form $xz \to b'$ if x is not a substring of a.*

Proof. Suppose that there are two rules $A = a \to yx$ and $B = xz \to b'$. If we assume that A does not feed B when x is not a substring of a, it follows that there are no strings in Σ^* of the shape yxz for which x is not a substring of a. Since, by definition, Σ^* contains all strings over the alphabet Σ, by the definition of the Kleene closure, this would be a contradiction $\square$

Lemma 5. *For all other rules A and B, A does not feed B.*

Proof. Assume that there are rules $A = a \to a'$ and $B = b \to b'$ s.t. a' is not ϵ and b is not a substring of a' and there is no string x such that x is a prefix of a' and a suffix of b or a suffix of a' and a prefix of b. Assume that A feeds B. This entails that the structural description of B is satisfied by some output of A. Since a' cannot have b as a substring, the structural description must be satisfied by the concatenation of a' and some other string x. If $a'x$ or xa' match the structural description of B (contains b as a substring) it follows that b must be a substring of x since, otherwise, it would have to overlap with a', contrary to stipulation. Thus, A cannot introduce a new substring meeting the structural description of B and therefore cannot feed B. $\square$

YW: Here is another solution for definition. From my perspective the logic is more clear and easy to build the rules for bleeding. But I am not sure whether it is correct

At the beginning we need to define the Rule Applicability.

Definition 2.4 (Rule Applicability)**.** For a rule $a \to b$, string s is rule applicable if and only if

$$a \in B(s)$$

Definition 2.5 (Feeding definition)**.** In phonology and historical linguistics, feeding order of phonological rules refers to a situation in which the application of a rule A creates new contexts in which a rule B can apply; it would not have been possible for rule B to apply otherwise. Formally with the definition of 2.4 we can define it as:

For rules $A : a \to a'$ and $B : b \to b'$, A feeds B if:

$$\exists s \quad \text{such that } b \in B(s) \wedge a' \in B(s) \wedge b \notin B(s.\text{replace}(a, a'))$$

We need some pre-definition for later assumption.

Definition 2.6 (Substring Prefix and Substring Suffix)**.** Let s, s' be strings over an alphabet Σ, where s is a substring of s'. We define the **Substring Prefix** $SP(s', s)$ and the **Substring Suffix** $SS(s', s)$ as follows:

$$\exists SP(s', s), SS(s', s) \in \Sigma^* \text{ such that } s' = SP(s', s) \cdot s \cdot SS(s', s). \tag{3}$$

Operation $A \cdot B$ is concatenation for string A and B.

Definition 2.7 (Maximum Common Substring)**.** Let a and b be two strings over an alphabet Σ. The **maximum common substring** of a and b is defined as:

$$MCS(a, b) = \arg \max_{s \in B(a) \cap B(b)} |s|$$

Here we give the formal definition of **context** rule B can be applied.

Definition 2.8 (Cascade Compatibility)**.** Let a, b be two strings over an alphabet Σ. We say that a and b are **Cascade Compatible**, denoted as $CC(a, b) =$ True if and only if there exists a maximum substring $i \in \Sigma^* = MCS(a, b)$ such that the corresponding Substring Prefix and Substring Suffix satisfy:

$$\begin{cases} SP(a, i) \in S(SP(b, i)) \quad \text{or} \quad SP(b, i) \in S(SP(a, i)), \\ SS(a, i) \in P(SS(b, i)) \quad \text{or} \quad SS(b, i) \in P(SS(a, i)). \end{cases} \tag{4}$$

Otherwise, we define $CC(a, b) =$ False, meaning a and b are **Cascade Incompatible**.

Lemma 6. *For string a and b, $CC(a, b) \iff \exists s$ such that $s = \hat{P} \cdot a \cdot \hat{S} = \bar{P} \cdot b \cdot \bar{S}$*

Proof. Sufficiency: If $CC(a, b)$, we denote:

$$SP = \arg \max_{s \in \{SP(a,i), SP(b,i)\}} |s|$$

$$SS = \arg \max_{s \in \{SS(a,i), SS(b,i)\}} |s|$$

Obvious, we can construct a string $s = SP \cdot i \cdot SS$ can be one string such that $s = \hat{P} \cdot a \cdot \hat{S} = \bar{P} \cdot b \cdot \bar{S}$

Necessity: If $\exists s$ such that $s = \hat{P} \cdot a \cdot \hat{S} = \bar{P} \cdot b \cdot \bar{S}$, it is clear that:

$$\hat{P} \cdot SP(a,i) = \bar{P} \cdot SP(b,i)$$

$$SS(a,i) \cdot \hat{S} = SS(b,i) \cdot \bar{S}$$

In this case,

$$\begin{cases} SP(a,i) \in S(SP(b,i)) \quad \text{or} \quad SP(b,i) \in S(SP(a,i)), \\ SS(a,i) \in P(SS(b,i)) \quad \text{or} \quad SS(b,i) \in P(SS(a,i)). \end{cases}$$

□

Definition 2.9 (Inverse Static Cascade Compatibility)**.** Let a, b, i be strings over Σ^*. We define the **Inverse Static Cascade Compatibility** of a and b with respect to i, denoted as $ISCC(a,b,i)$, if $\exists SP(a,i), SS(a,i)$ such that $i \notin B(a) \cap B(b)$ otherwise:

$$\begin{cases} SP(a,i) \notin S(SP(b,i)) \quad \text{and} \quad SP(b,i) \notin S(SP(a,i)), \\ SS(a,i) \notin P(SS(b,i)) \quad \text{and} \quad SS(b,i) \notin P(SS(a,i)). \end{cases}$$

Based on Definition 2.8, 2.9 and Lemma 6, we can write the algorithm as:

Definition 2.10 (Feeding)**.** For the following two rules, rule $b \to b'$ can feed rule $a \to a'$ if and only if

$$f(a \to a', b \to b') = \begin{cases} \top, & CC(a', b) \land ISCC(a, b, MCS(a', b)) \\ \bot, & \text{otherwise.} \end{cases}$$

Proof. Necessity: Based on Lemma 6, we can construct a string $s = SP \cdot i \cdot SS$ which is the shortest string that satisfies lemma 6. In this case $b \in B(s) \land a' \in B(s)$. Also, the string after replacement should be $\{x | x \in (a, a \cdot SS, SP \cdot a, SP \cdot a \cdot SS)\}$. The ISCC can make sure that $b \notin B(x)$.

Sufficiency: Vice versa. □

YW: alg not complete now

Algorithm 1 Checking if A feeds B

Require: Rule $A : a \to a'$, Rule $B : b \to b'$
Require: Helper functions: $B(\cdot)$, $P(\cdot)$, $S(\cdot)$, $E(\cdot,\cdot)$, $MCS(\cdot,\cdot)$
Ensure: $f(A, B) =$ True if A feeds B, otherwise False

$i \leftarrow MCS(a', b)$
$SP(a', i), SS(a', i) \leftarrow E(a', i)$
$SP(b, i), SS(b, i) \leftarrow E(b, i)$
if $SP(a', i) \notin S(SP(b, i))$ **and** $SP(b, i) \notin S(SP(a', i))$ **then**
  return False
end if
if $SS(a', i) \notin P(SS(b, i))$ **and** $SS(b, i) \notin P(SS(a', i))$ **then**
  return False
end if
if $i \notin B(a)$ **then**
  return True
end if
$SP(a, i), SS(a, i) \leftarrow E(a, i)$
if $SP(a, i) \in S(SP(b, i))$ **or** $SP(b, i) \in S(SP(a', i))$ **then**
  return False
end if
return True

3 Methodology

3.1 Dataset Creation

We construct our benchmark by automatically generating synthetic input-output examples involving non-trivial string transformations over short character sequences. Each example consists of five input strings, five corresponding output strings, a sequence of transformation rules, and a graph that encodes the interactions among these rules.

3.1.1 Input Sampling

Each input string is sampled independently, with a length uniformly drawn between 2 and 6 characters. Characters are sampled uniformly from a restricted alphabet (e.g., `abcdefghijkxyz`). At this stage, the strings are unaltered; however, we ensure they are valid for subsequent transformation steps.

3.1.2 Program Sampling and Output Generation

We then sample a sequence of string transformation rules. Each rule replaces the first occurrence of a randomly chosen substring (1–3 characters) with another randomly selected substring, both drawn from the vocabulary. The rules are applied sequentially to each input string, modifying the list step by step.

To ensure that meaningful transformations occur, we employ rejection sampling: if the output strings are identical to the inputs, the sample is discarded. This guarantees that every retained example involves at least one substantive transformation.

3.1.3 Rule Interactions: BFCC DAGs

In addition to the transformation process, we annotate each example with a directed acyclic graph (DAG) that represents interactions among rules in the sequence. We refer to this as the **BFCC DAG**, capturing four linguistically inspired interaction types:

- **Feeding:** Rule A creates substrings that enable Rule B to apply.
- **Bleeding:** Rule A removes substrings that Rule B requires.
- **Counterfeeding:** Rule A could have enabled Rule B, but B precedes A.
- **Counterbleeding:** Rule A could have blocked Rule B, but B precedes A.

We infer these interactions by analyzing how rules modify substrings and affect one anothers applicability. The resulting graph encodes the temporal and structural dynamics of the rule sequence beyond individual rule effects.

3.1.4 Dataset Balancing

To ensure diversity across transformation types and interaction patterns, we apply rejection sampling once more. Each example is labeled with a 4-bit vector indicating the presence of the four BFCC interaction types. We track the distribution of examples across interaction patterns and transformation sequence lengths, retaining only those that contribute to underrepresented categories.

Examples that do not improve this coverage are discarded. We also monitor sampling efficiency and terminate early if the process becomes too inefficient.

This procedure yields a dataset that is balanced both in the complexity of transformations and the diversity of rule interactions, providing a rich testbed for models requiring compositional and structural reasoning.

3.2 Analysis of Dataset

Model	**Pass@1**	**Edit Similarity**
Codestral-22B	0.1552	0.1758
Qwen2.5-32B-Instruct	0.2006	0.2277
Qwen2.5Coder-32B-Instruct	0.2089	0.2359
QwQ-32B	0.177	0.2231
Qwen3-32B	0.1784	0.1818
Qwen3-30B-A3B	0.2913	0.3
DeepSeek-R1-Distill-Qwen-32B	0.1562	0.1254
DeepSeek-R1	0.321	0.3707
o3-mini	0.0464	0.0533
o4-mini	0.3871	0.4154
Gemini 2.0 Flash Thinking Exp 01-21	0.4622	0.522
Claude-3.5-Sonnet	0.2732	0.3416
Claude-3.7-Sonnet	**0.5111**	**0.5647**

Table 1: We compute the pass@1 and

4 Experiments

4.1 Results

4.2 Discussion

5 Submission of papers to NeurIPS 2024

Please read the instructions below carefully and follow them faithfully.

5.1 Style

Papers to be submitted to NeurIPS 2024 must be prepared according to the instructions presented here. Papers may only be up to **nine** pages long, including figures. Additional pages *containing only acknowledgments and references* are allowed. Papers that exceed the page limit will not be reviewed, or in any other way considered for presentation at the conference.

The margins in 2024 are the same as those in previous years.

Authors are required to use the NeurIPS LaTeX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for rejection.

5.2 Retrieval of style files

The style files for NeurIPS and other conference information are available on the website at

`http://www.neurips.cc/`

The file `neurips_2024.pdf` contains these instructions and illustrates the various formatting requirements your NeurIPS paper must satisfy.

The only supported style file for NeurIPS 2024 is `neurips_2024.sty`, rewritten for LaTeX 2_{ε}. **Previous style files for LaTeX 2.09, Microsoft Word, and RTF are no longer supported!**

The LaTeX style file contains three optional arguments: `final`, which creates a camera-ready copy, `preprint`, which creates a preprint for submission to, e.g., arXiv, and `nonatbib`, which will not load the `natbib` package for you in case of package clash.

Preprint option If you wish to post a preprint of your work online, e.g., on arXiv, using the NeurIPS style, please use the `preprint` option. This will create a nonanonymized version of your work with the text "Preprint. Work in progress." in the footer. This version may be distributed as you see fit, as long as you do not say which conference it was submitted to. Please **do not** use the `final` option, which should **only** be used for papers accepted to NeurIPS.

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The file `neurips_2024.tex` may be used as a "shell" for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in Sections 6, 7, and 8 below.

6 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by ½ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow ¼ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors' names are set in boldface, and each name is centered above the corresponding address. The lead author's name is to be listed first (left-most), and the co-authors' names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section 8 regarding figures, tables, acknowledgments, and references.

7 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

7.1 Headings: second level

Second-level headings should be in 10-point type.

7.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

8 Citations, figures, tables, references

These instructions apply to everyone.

8.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

Hasselmo, et al. (1995) investigated...

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2024` package:

```
\PassOptionsToPackage{options}{natbib}
```

If `natbib` clashes with another package you load, you can add the optional argument `nonatbib` when loading the style file:

```
\usepackage[nonatbib]{neurips_2024}
```

As submission is double blind, refer to your own published work in the third person. That is, use "In the previous work of Jones et al. [4]," not "In our previous work [4]." If you cite your other papers that are not widely available (e.g., a journal paper under review), use anonymous author names in the citation, e.g., an author of the form "A. Anonymous" and include a copy of the anonymized paper in the supplementary material.

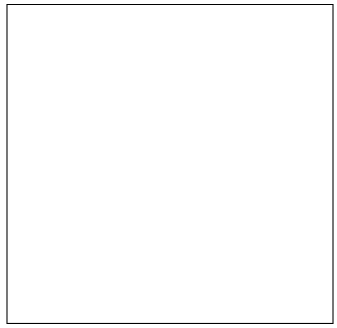

Figure 1: Sample figure caption.

Table 2: Sample table title

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

8.2 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number[1] in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.[2]

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All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

8.4 Tables

All tables must be centered, neat, clean and legible. The table number and title always appear before the table. See Table 2.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules.* We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

`https://www.ctan.org/pkg/booktabs`

This package was used to typeset Table 2.

[1] Sample of the first footnote.
[2] As in this example.

8.5 Math

Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You really shouldn't be using $$ anyway; see `https://tex.stackexchange.com/questions/503/why-is-preferable-to` and `https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath` for more information.)

8.6 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the **References** section; see below). Please note that pages should be numbered.

9 Preparing PDF files

Please prepare submission files with paper size "US Letter," and not, for example, "A4."

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF files uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdffonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

  Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

9.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (`http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf`)

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[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

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[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

A Appendix / supplemental material

Optionally include supplemental material (complete proofs, additional experiments and plots) in appendix. All such materials **SHOULD be included in the main submission.**

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